

Vishal Nayak

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SUMMARY

Experienced Data Scientist (**3.5+ years**) specializing in **Natural Language Processing (NLP)**, **Statistical Machine learning and Deep Learning**. Proficient in **Python, PyTorch, and TensorFlow** for developing **advanced NLP models, sentiment analysis, named entity recognition, and language generation**. Skilled in data preprocessing, feature engineering, model optimization, and Model deployment on AWS cloud.

EDUCATION

Indian Institute of Information Technology (IIIT Gwalior)

Post Graduate in Information Technology (IPG - 5 yrs)

Gwalior, Madhya Pradesh, India

July 2015 – Aug 2020

Kendriya Vidyalaya, Raigunj

Senior Secondary (CBSE) - Computer Science

Raigunj, West Bengal, India

Mar 2012 – April 2013

TECHNICAL SKILLS

Expertise Areas	: ML (Regression/Classification), Deep Learning (CNN/RNN/LSTM/Word2vec/Glove), NLP (Encoder-Decoder/ Transformers / BERT / T5 / LLMs), Vector Embedding Search (Qdrant), Sentence Embedding
Languages	: Python, C/C++, HTML, CSS
Frameworks	: PyTorch, TensorFlow, Keras, Scikit-learn, Django, Flask, Fast API
Libraries	: NLTK, SpaCy, Genism, Transformers (Hugging Face), Matplotlib, Pandas, Numpy
Dev Tools	: PyCharm, Jupyter Notebook, Git, Gitlab, Visual Studio
Other Skills	: ElasticSearch, AWS / Azure cloud, AWS Sagemaker, S3, MLOPs, Docker, REST API, Statistics, Probability, Data Modeling, Data Analysis, Data Visualisation, Model deployment, Model Optimization, Generative AI, LLM, Prompt Engineering, Language models, ONNX, TorchServe, SQL, Microsoft Office, Microsoft Excel

EXPERIENCE

Data Scientist

Foundit.ai (Job Search)

Sep 2023 – Present

Onsite – Bangalore, Karnataka, India

- Built in-house **Resume Parser** API to replace 3rd party Sovren API to reduce costs.
- Used **OpenAI GPT-4** model to parse resumes and extract required fields during candidate registrations.
- Trained and implemented our own **Spacy NER** model to extract the necessary fields with **>85% accuracy**, replacing the previous **OpenAI** solution.
- Matched resume parse data with our core master database data to prevent storing duplicate information in the master database using **vector embedding, fuzzy matching**, and **TF-IDF** vectorizer approach for fields like Designation, Company, College, Highest Qualification, and Locations.
- Developed a **character-based LSTM** model for predicting gender from names specific to India, Gulf, and SEA regions.
- Collected and preprocessed a diverse dataset of names with corresponding gender labels. Trained the **LSTM** model using dropout regularization and hyperparameter tuning to achieve optimal performance. Achieved **>90% accuracy** in gender prediction during model evaluation and testing.

Machine Learning Engineer

Turtlemint Private Limited (Insurance Company)

Feb 2022 – Sep 2023

Onsite – Pune, Maharashtra, India

- Led and guided team to develop end-to-end product for **Data extraction** from diverse policy documents, including Motors, health, life, RC copies, and Aadhaar cards.
- Developed end to end data extraction pipeline using:
 - * Built **classification model** to identify important pages from policy PDFs using image conversion and text preprocessing.
 - * Utilized **SVM**-based text classification with keyword analysis for page importance.
 - * Used **AWS Textract OCR** service to extract text from important pages.

- * Developed in-house **annotations tool** for **NER** model data preparation.
- * Trained **LayoutLM Transformer**-based **NER** model achieving over **95% accuracy**.
- Addressed challenges, overcoming **512-token** constraint with **LayoutLM** model by implementing stride window approach for full-page field extraction.
- Deployed model using **Torch** Serve and contributed to automated model training and deployment platform with **AWS SageMaker** pipeline integration.

Data Scientist

Knowdis AI Private Limited

Sep 2020 – Feb 2022

Remote - New Delhi, Delhi, India

- Contributed to a **10-20% improvement** in IndiaMart search results, optimizing the user experience for Indian customers.
- Achieved over **85% accuracy** in language detection (English vs. Roman-Hindi) by fine-tuning **BERT** model with 1 lakh search queries.
- Developed Roman Hindi to English Translator model by fine tuning **Google Multilingual T5 Transformer** model on IndiaMart search queries.
- Developed model for Email Spam detector by fine-tuning **BERT** model using 1.5 lakh internet-sourced data and achieved over **90% accuracy**.
- Deployed PyTorch model on **AWS** using Torch Serve to provide efficient and scalable model serving for real-time applications.

PROJECTS

Project 1

Logistics Regression, SMOTE

- Utilized machine learning techniques to detect fraud transactions from historical data, minimizing financial losses.
- Utilized a credit card transaction dataset containing numerical features derived from PCA transformation.
- Features included Time, Amount, and transformed data from PCA.
- Target variable: Binary classification of fraud (1) and non-fraud (0) transactions.
- Applied classification algorithms: Logistic Regression, K-Nearest Neighbor, Support Vector Machine, Decision Tree.
- Compared techniques for handling imbalanced dataset: Random-Under Sampling and SMOTE (Synthetic Minority Oversampling Technique).
- Evaluated models using Precision-Recall and ROC curves.

ACHIEVEMENTS

- Secured rank under Top 5 in Gartner Data science Hackathon by Hackerearth, 2018.
- Secured rank under Top 20 in Data science Hackathon by Analytics Vidya.
- Secured Runner up in Twaran 2k17 (Inter - IIIT Sports fest)

CERTIFICATIONS

- Python for Data Science - DataCamp
- Machine Learning Specialization - By Andrew Ng
- Deep Learning Specialization - By Andrew Ng
- NLP with Deep Learning - Stanford University

INTERESTS

- Cricket/Basketball
- Travelling
- Watching Web series
- Learning New Technologies